

Principal Component Analysis: From First Principles

PCA is a dimensionality reduction technique. It transforms many correlated features into fewer new features called principal components.

Why PCA Is Needed

Datasets can have many features. Some features may carry overlapping information. PCA compresses the data while preserving as much variation as possible.

Principal Components

A principal component is a new direction in the data. The first principal component captures the most variance. The second captures the next most variance while being perpendicular to the first.

Variance

PCA tries to preserve variance because high-variance directions often contain important structure.

Core Transformation

PCA projects data onto new axes:

$$Z = XW$$

Here, X is the scaled data, W contains principal component directions, and Z is the transformed lower-dimensional data.

Explained Variance

Explained variance tells us how much information each principal component keeps. Cumulative explained variance helps choose the number of components.

When to Use It

Use PCA for visualization, noise reduction, compression, and reducing correlated features. Avoid it when interpretability of original features is essential.